

Software Requirements Specification (SRS)

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I. Introduction

Chess is a highly strategic game that requires logical thinking, prediction, and planning. Developing artificial intelligence (AI) that can play chess at a high level has become an important research topic in the field of artificial intelligence. In recent years, reinforcement learning (RL) has proven to be highly effective in training AI to play chess without relying on opening libraries or pre-existing training data.

Reinforcement learning is a branch of machine learning in which an agent learns how to act in an environment to maximize its rewards. In chess AI, this method allows the system to learn through trial and error, improving its strategy over time without human guidance. Achievements such as DeepMind's AlphaZero have demonstrated that AI can achieve superhuman levels simply by playing against itself.

This research will focus on exploring popular reinforcement learning algorithms, analyzing how they are applied to train chess-playing AI, comparing them with traditional methods, and evaluating their performance. The results from the research could open up new directions in the development of chess-playing AI as well as broader applications in other areas such as optimization, planning, and decision making.

1.1. Purpose

The objective of this study is to explore and evaluate the Reinforcement Learning (RL) method in training artificial intelligence (AI) to play chess, in order to better understand the effectiveness, advantages and limitations of this method compared to traditional approaches. Specifically, the study aims at the following objectives:

- Understanding reinforcement learning algorithms

- Analyzing popular RL algorithms applied in AI playing chess, including Deep Q-Network (DQN), Monte Carlo Tree Search (MCTS), AlphaZero, MuZero...

- Evaluating the operating mechanism, learning ability and strategy optimization of these algorithms.

- Comparing with traditional methods

- Comparing the performance of reinforcement learning AI with chess engines using traditional algorithms such as Stockfish (based on Minimax, Alpha-Beta Pruning) or Leela Chess Zero (Lc0 - using deep neural networks combined with MCTS).

- Evaluate the differences in decision-making, adaptability, and playing performance of each method.

- Build and train a chess-playing AI model

- Design and train a chess-playing AI model using RL.

- Analyze the training results, learning speed, playing strategies, and performance against other engines.

Application and extension

Evaluate the potential applications of reinforcement learning in areas other than chess, such as planning, strategy optimization, and decision making in complex environments.

Suggest future directions for developing and improving chess-playing AI.

1.2 Intended Audience

This study targets the following audiences, including different game modes in chess AI:

Chess-playing AI

AI models using Reinforcement Learning (RL) to play chess.

Popular algorithms such as Deep Q-Network (DQN), Monte Carlo Tree Search (MCTS), AlphaZero, MuZero.

Traditional chess engines such as Stockfish, Leela Chess Zero (Lc0) for performance comparison.

Reinforcement Learning Methods in AI Training

Building and optimizing chess-playing AI through self-play, training with available data or training from simulated environments.

Improving AI prediction and planning capabilities through MCTS combined with Deep Neural Networks (DNNs).

Game modes in AI chess

Play offline: Players can play against AI without an internet connection.

Play with AI: AI will play as an opponent, with many levels from basic to advanced.

Play online: Support playing against other players via the internet, can integrate rankings and level assessment systems (Elo rating).

Practice mode: AI can suggest moves, analyze games and help players improve their skills.

Training environment and data

8x8 chessboard, chess pieces and chess rules according to international standards (FIDE).

Technology and framework used: OpenAI Gym, TensorFlow, PyTorch, RLlib to train and deploy AI.

Input data: Chess games from databases (FICS, Lichess, Chess.com) or AI plays against itself to collect data.

Applications and Research Subjects

AI researchers, engineers, students in Computer Science, Artificial Intelligence, Data Science.

Chess enthusiasts who want to experience and learn from AI.

RL applications are not only in chess but also in other strategy games, decision optimization, and automated planning.

1.3. Scope

The chess game website will be a web-based platform where users can:

- Register and log in to their account, allowing them to create a personal profile and save their game progress.
- Play chess against other users in real-time or against AI opponents with adjustable difficulty settings to suit different skill levels.
- Join ranked matches and leaderboards to compete against other players globally and track their performance over time.
- Engage in real-time chat with other players during matches, fostering a sense of community and interaction within the platform.
- View match history and game statistics, allowing users to analyze their gameplay, track their win/loss ratio, and improve their strategy.
- Access the site seamlessly across both desktop and mobile, ensuring a responsive and adaptive user experience regardless of the device being used.
- Receive notifications for challenges, game invites, and rank updates to stay up to date with ongoing matches and community activities.
- Customize their gaming experience with different themes, board styles, and UI options for a more personalized and engaging experience.

1.4. Definitions

Chess: A classic two-player strategy board game played on an 8x8 grid, where each player controls 16 pieces (king, queen, rooks, bishops, knights, and pawns). The objective of the game is to checkmate the opponent's king by placing it under an inescapable threat. Chess is a globally recognized game with various rule sets, including classical, rapid, blitz, and bullet formats.

AI (Artificial Intelligence): A computational system designed to simulate human-like decision-making and problem-solving abilities when playing against users. The AI opponent in the Chess game will have varying difficulty levels, employing algorithms such as Minimax with Alpha-Beta Pruning or Monte Carlo Tree Search (MCTS) to evaluate board positions and make optimal moves based on strategic considerations. Additionally, advanced chess engines like Stockfish or Leela Chess Zero (Lc0) may be integrated for a more challenging and realistic gameplay experience.

UI/UX (User Interface and User Experience):

UI (User Interface): Refers to the design and layout of graphical elements that users interact with, such as chess pieces, game board, buttons, and menus.

UX (User Experience): Encompasses the overall experience a user has while navigating the website, ensuring ease of use, efficiency, and engagement. A well-designed UI/UX will provide intuitive controls, visually appealing aesthetics, and smooth interactions, enhancing the player's gaming experience across different devices.

II. General Description

2.1. Product Perspective

The Chess Game Website will be a standalone web application designed to provide an engaging and interactive online chess experience. It will feature both multiplayer and AI-based gameplay modes, allowing users to challenge other players in real-time or test their skills against computer opponents with varying levels of difficulty.

The platform will integrate a robust user authentication system to ensure secure access, allowing users to create and manage their accounts, track their performance, and interact with the community. Real-time communication capabilities will be implemented, enabling players to chat with their opponents, receive notifications of game invites, and interact through messaging features.

The game logic will be structured to ensure fair play, with chess rules strictly enforced to maintain the integrity of matches. Additionally, the platform will be optimized for performance, ensuring low-latency interactions and seamless gameplay across both desktop and mobile devices.

Future enhancements may include:

Tournament features for competitive play.

Advanced AI enhancements for stronger gameplay challenges.

Expanded social functionality to further enrich the user experience.

2.2. User Classes and Characteristics

Guest Users:

Guest users are individuals who visit the site without creating an account. They can play practice games against AI or other guest users without registering. However, they do not have access to ranking matches, leaderboards, or chat functionality. Their gameplay progress is not saved, and they cannot view match history or statistics. The primary purpose of guest users is to allow new visitors to experience the game before registering an account.

Registered Players:

Registered players have full access to all essential features of the platform. They can:

Join competitive matches with other players.

Engage in real-time chat with opponents.

View match history and performance statistics.

Customize their profile and track ranking on the leaderboard.

Receive notifications about challenges and game invites.

Adjust game settings and choose AI difficulty levels.

Participate in tournaments if implemented in future updates.

Admins:

Admins are responsible for overseeing the platform and ensuring a fair and smooth gaming experience. Their include tasks:

Managing user accounts and monitoring ongoing/completed matches.

Moderating chat to prevent inappropriate behavior.

Handling reports of misconduct and enforcing penalties such as temporary or permanent bans.

Overseeing site maintenance, including system performance metrics, server status, and security logs.

Ensuring platform stability and data integrity.

2.3. Operating Environment

The Chess Game Website will be a fully web-based application, eliminating the need for users to download or install any software. It will be accessible through modern web browsers, including Google Chrome, Mozilla Firefox, Safari, and Microsoft Edge, ensuring compatibility across different platforms.

The platform will feature a responsive design that dynamically adjusts its layout and interface elements to provide an optimal viewing and interaction experience on desktops, tablets, and mobile devices. This ensures usability across different screen sizes and resolutions, offering a consistent and intuitive experience for all users.

The game and associated data will be hosted on cloud-based or dedicated servers to ensure reliability, scalability, and high availability. The cloud infrastructure will allow the system to efficiently manage traffic spikes, support multiple concurrent users, and provide a seamless gaming experience without performance issues.

Security measures such as SSL encryption, firewall protection, and regular security audits will be implemented to secure user data and prevent unauthorized access or cyber threats. The system will also support data backups to prevent loss of game history and account information.

2.4. Constraints

The system must support real-time game updates with minimal latency, ensuring smooth gameplay and preventing delays in move execution during matches.

User data must be securely stored and handled, implementing encryption and authentication protocols to prevent unauthorized access and data restricted.

The AI opponent should feature multiple difficulty levels, catering to both beginner and advanced players. The AI should progressively challenge users by making strategic moves based on advanced algorithms, adapting its playstyle according to the player's skill level.

The system must be designed to handle a high number of concurrent users efficiently, ensuring performance remains optimal even during peak usage times.

III. Specific Requirements

3.1. Functional Requirements

3.1.1. User Registration and Login

Users must be able to create an account to access all features of the Chess Game Website, ensuring a personalized experience tailored to their preferences and gameplay history. The registration process is a crucial first step for new users, as it allows them to engage with the platform fully, including:

Participating in online matches (against AI or real players).

Accessing and managing their profiles.

Tracking their performance and ranking over time.

Saving game statistics and reviewing match history.

Interacting with other players through chat and notifications.

By creating an account, users can enjoy these features, which enhance the overall gaming experience.

During the registration process, users are required to provide several key pieces of information:

Email : Used for account verification, password recovery, and important notifications.

Password: A secure password that protects the user's account and personal information.

Home Register | Login

Register

Email

Enter your email

Note: Please use your own email, otherwise you will not be able to change your password later.

Password

Enter your password

Confirm Password

Confirm your password

First Name

Enter your first name

Last Name

Enter your last name

Birthday

mm/dd/yyyy

Register

Figure 1: Register

Users should be able to securely log in and out of their accounts to ensure their personal data and game progress are protected. This functionality is crucial to maintaining user privacy and security, as it allows users to access their profiles, track their game statistics, and participate in online matches while protecting their accounts from unauthorized access. The login process should be simple and intuitive, allowing users to enter their login credentials with ease. Once logged in, users should be given immediate access to a personalized dashboard where they can view their game history, current matches, and any notifications related to their account. Additionally, the logout feature is equally important, as it allows users to safely exit their session, especially when using a shared or public device, minimizing the risk of others accessing their account. When logging into their account, users are required to provide two essential pieces of information:

Email: This is a unique identifier for each user on the platform, allowing the system to identify and authenticate the individual.

Password: The secure password that the user set earlier during the registration process. This password is very important to ensure the security of the user's account.

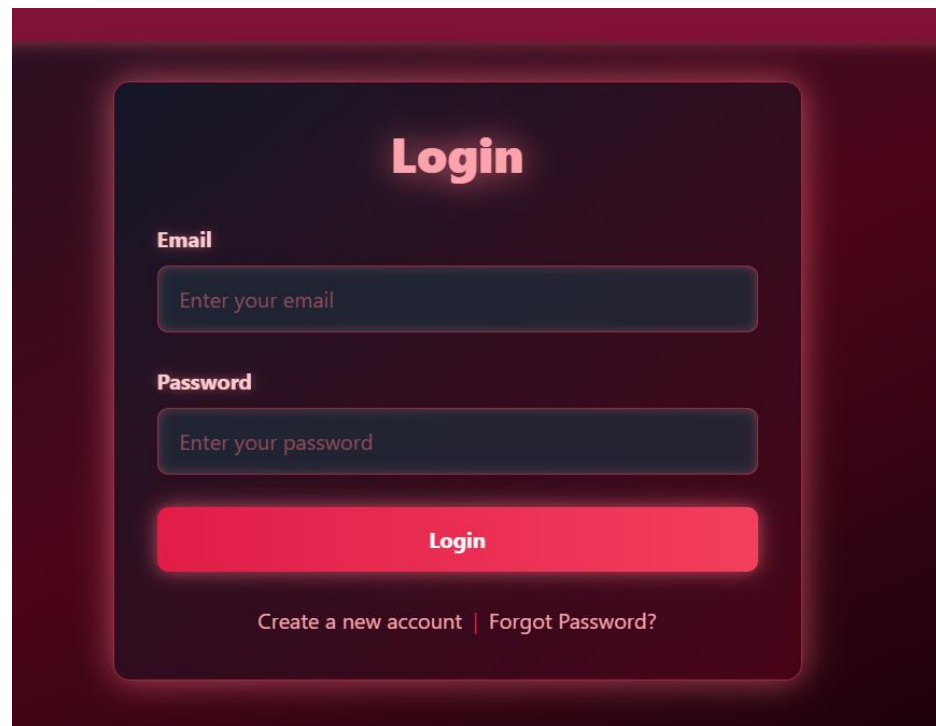


Figure 2:Login

3.1.2. Game Mechanics

Single Player Mode (Against AI)

In single player mode, users can play against Artificial Intelligence (AI) opponents, allowing them to enjoy Chess even when there are no other players. This mode is beneficial for:

New players who want to practice and learn chess strategies.

Experienced players who want to improve their skills without the pressure of competing against real opponents.

The AI is designed to simulate different difficulty levels, providing an experience that is tailored to the user's skill level. The AI can use Minimax with Alpha-Beta Pruning or Monte Carlo Tree Search (MCTS), and advanced engines such as Stockfish or Leela Chess Zero can be integrated for more powerful gameplay.

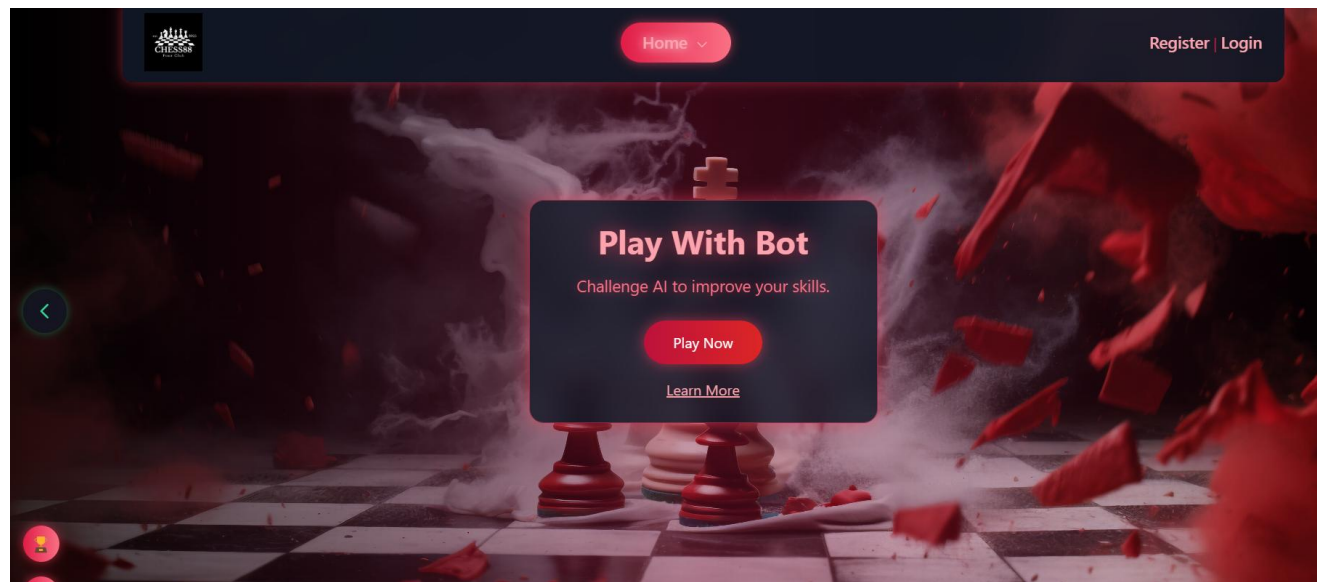


Figure 3: Play with Bots

Online Multiplayer Mode

In online mode, users can:

Play real-time chess matches with other players around the world.

Join ranked matches to compete on the leaderboards.

Play casual games without affecting your ranking.

Challenge your friends by sending them direct game invites.

Engage in real-time chat with opponents during matches.

Receive game invite notifications, ranking updates, and tournament notifications.

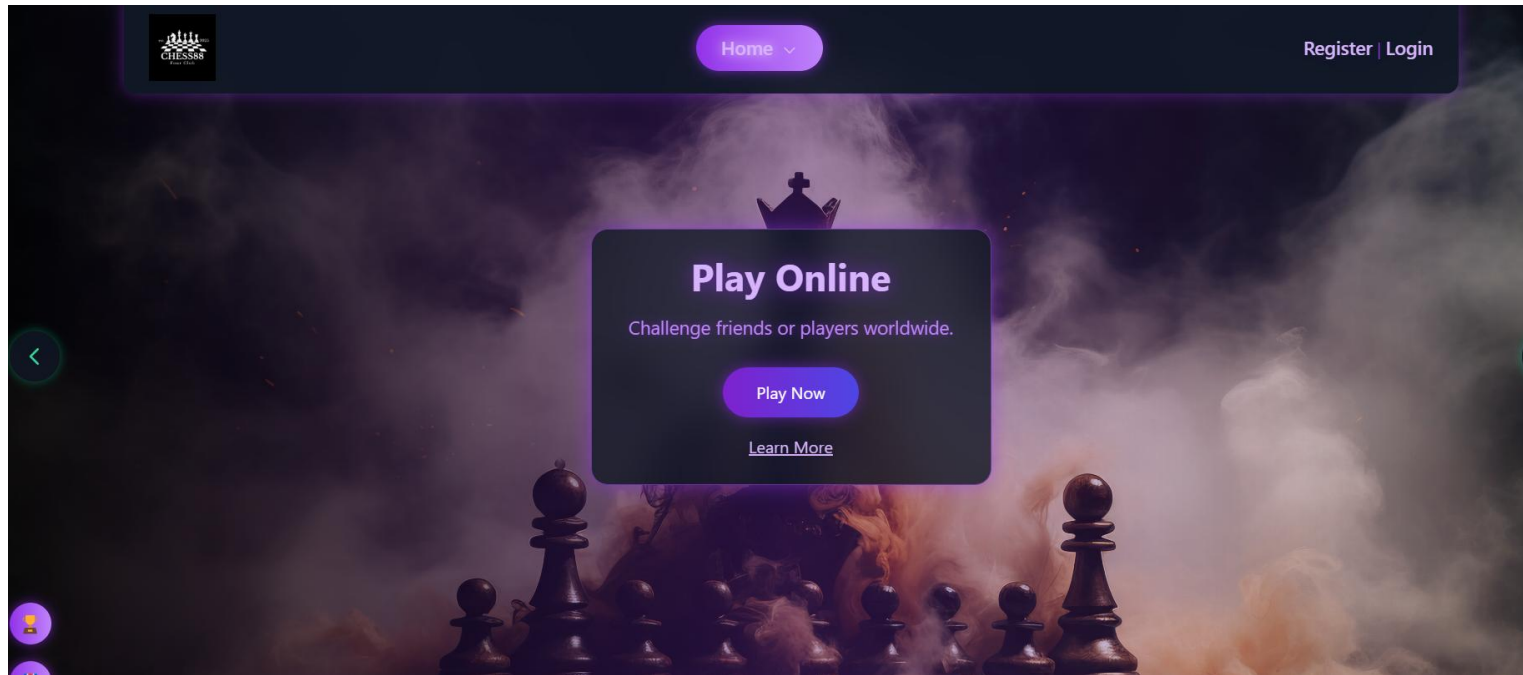


Figure 4: Play online

Offline Mode

In offline mode, users can:

Play chess against AI without an internet connection.

Play two-player games locally on the same device.

Save their match history and continue unfinished games when they are back online.

This mode is ideal for players who want to enjoy chess without an internet connection.

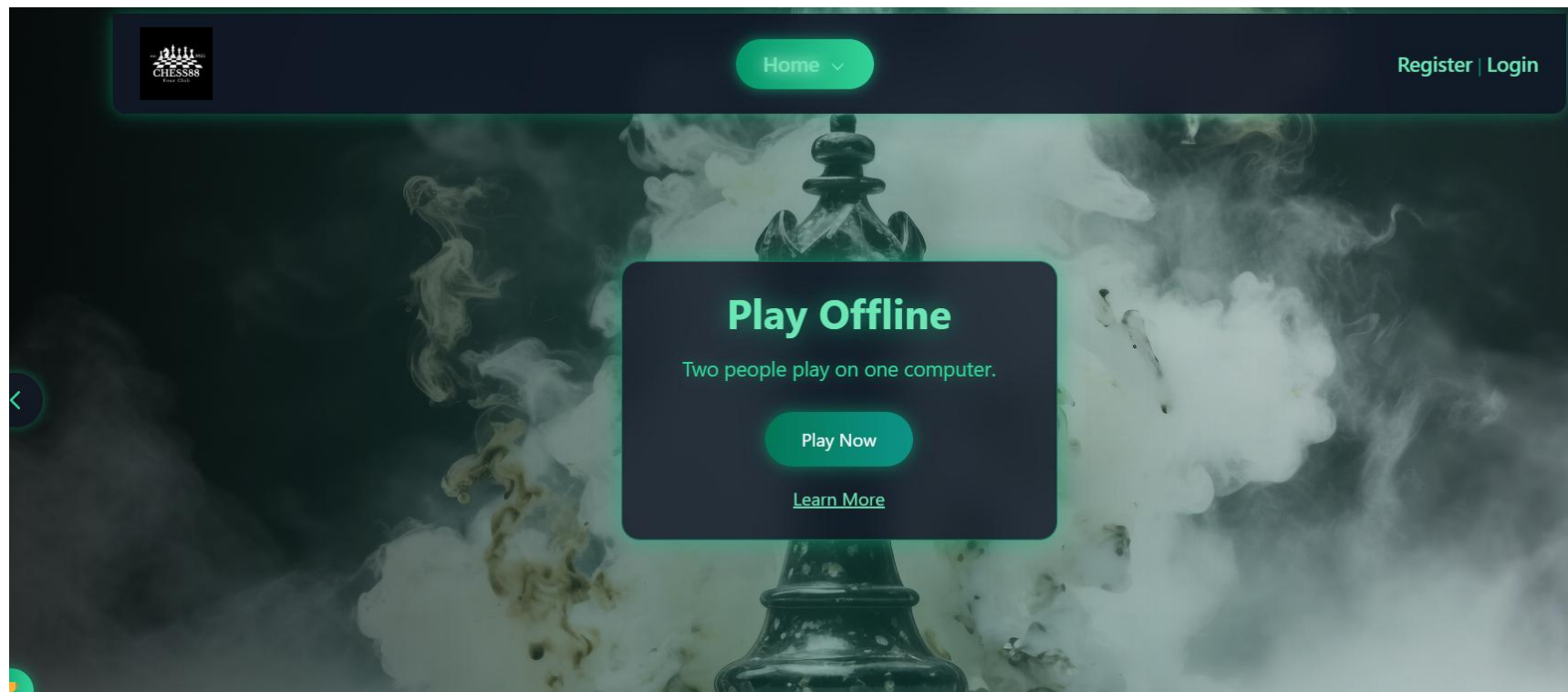


Figure 5: Play offline

3.1.3.Game board.

After the player presses the "Play" button, they will be transferred to the game room, where the chess board interface appears. The standard chess board is 8x8 in size, consisting of alternating white and black squares, marked from A to H in columns and 1 to 8 in rows for easy identification of the board. Each player controls 16 chess pieces, including 1 King, 1 Queen, 2 Rooks, 2 Bishops, 2 Knights and 8 Pawns. The white army always goes first according to international chess rules.

The interface chess board is designed to be intuitive, friendly and adaptive on all devices, from computers to phones. Players can choose the classic 2D or modern 3D interface, and switch between day and night ban modes for an optimal experience. The system supports valid move display, last move highlight, automatic chess board rotation with player color options, and battle history recording in PGN format.

Players can move chess pieces by drag & drop or direct selection. In addition, the chess board also supports chess notes, helping players mark important positions for tactical analysis. All these features ensure a smooth, convenient, and suitable online and offline chess playing experience for all levels of players.



Figure 6:Chessboard interface

During gameplay, players interact with the board by clicking on the army flag they want to move, then selecting the square to make the move. The input process is designed to be intuitive and easy to use, helping players perform moves quickly and accurately. When a player selects a flag, the system will highlight the squares that the flag can move, making it easy for them to determine the validity of their choices.

After making a move, the system will provide visual feedback, giving ideas such as highlighting the move just made or displaying the effect of the chess piece moving, helping players easily follow the progress of the battle. If the move is invalid, the system will display a warning to avoid confusion. This input method helps optimize the chess playing experience, ensuring a smooth finish speed and supporting players of all levels

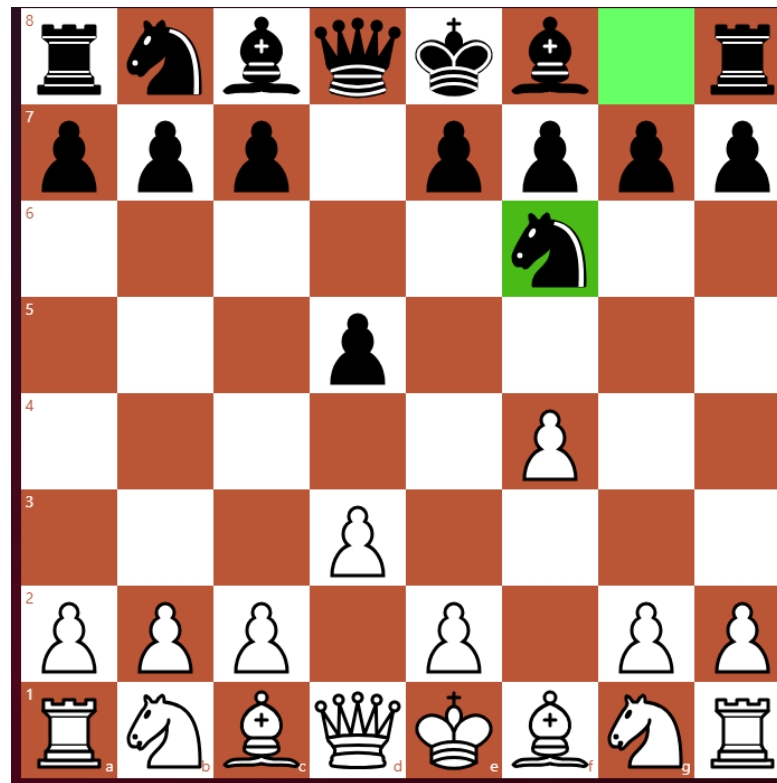


Figure 7:Chessboard interface 2

3.1.4.Rules of the Game

1. Chessboard and Pieces

The chessboard has 64 squares (8x8) with two alternating colors (black - white).

Each player has 16 pieces: 1 King, 1 Queen, 2 Rooks, 2 Bishops, 2 Knights and 8 Pawns.

Arrange the pieces:

Row 1 (white) and row 8 (black) in the following order: Rook - Knight - Bishop - Queen - King - Bishop - Knight - Rook.

The Queen is placed on the square of the same color (white Queen on white square, black Queen on black square).

Row 2 and row 7 are Pawns.

2. How to move the pieces

King: Move one square in any direction (horizontally, vertically, diagonally).

Queen: Move horizontally, vertically or diagonally, unlimited squares.

Rook: Move horizontally or vertically, unlimited squares.

Bishop: Move diagonally, unlimited squares.

Knight: Moves in an "L" shape (2 squares in one direction and 1 square perpendicular). Knight can jump over other pieces.

Pawn:

Moves 1 square straight, but the first move can move 2 squares.

Capture diagonally.

If a Pawn reaches the end rank, it can be promoted to a Queen, Rook, Bishop, or Knight.

3. Captures and special rules

Capture: When a piece enters a square occupied by an opponent's piece, the opponent's piece is removed from the board.

Castling: A special move between the King and Rook to protect the King and activate the Rook. Conditions:

The King and Rook have not moved.

There are no pieces between the King and Rook.

The King is not in check or passing through a controlled square.

En passant: When a pawn moves 2 squares from its starting position and stops next to an opponent's pawn, the opponent can capture it as if it had only moved 1 square.

Check and Checkmate:

Check: When the King is attacked.

Checkmate: When the King is in check and has no legal move to escape => The game ends.

4. End of the game

Checkmate: One side loses when the King is in check and cannot escape.

Draw is possible if:

There are no legal moves but no check (Stalemate).

Both sides agree to draw.

Not enough pieces to checkmate (Example: King vs King, King + Knight vs King).

Repeat the same position three times.

No pieces are captured and no pawns have moved in 50 consecutive moves.

The condition for a player to win the game is that the King is in check and has no way to escape, then he will win.

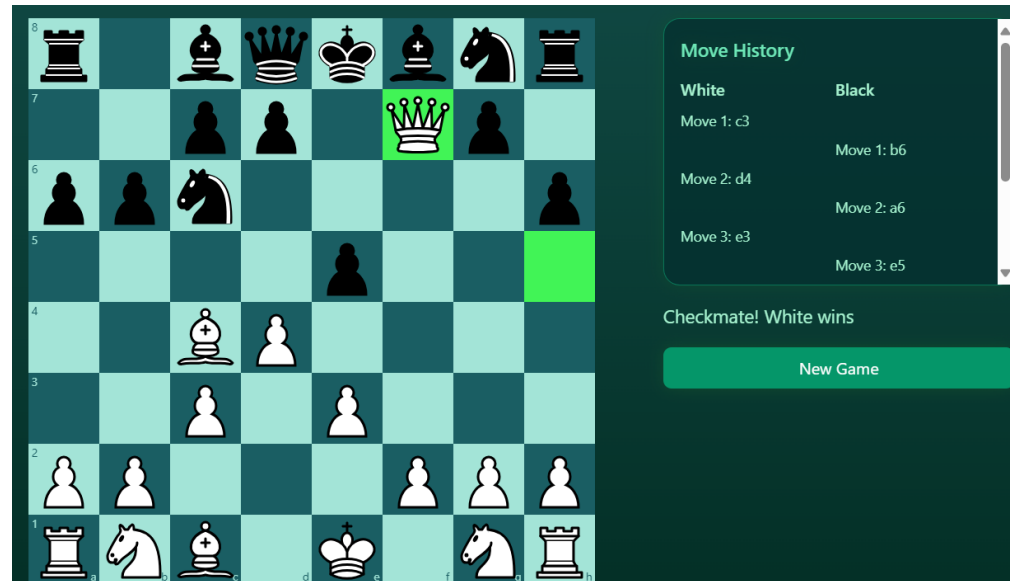


Figure 8:Win chess

Draw the game

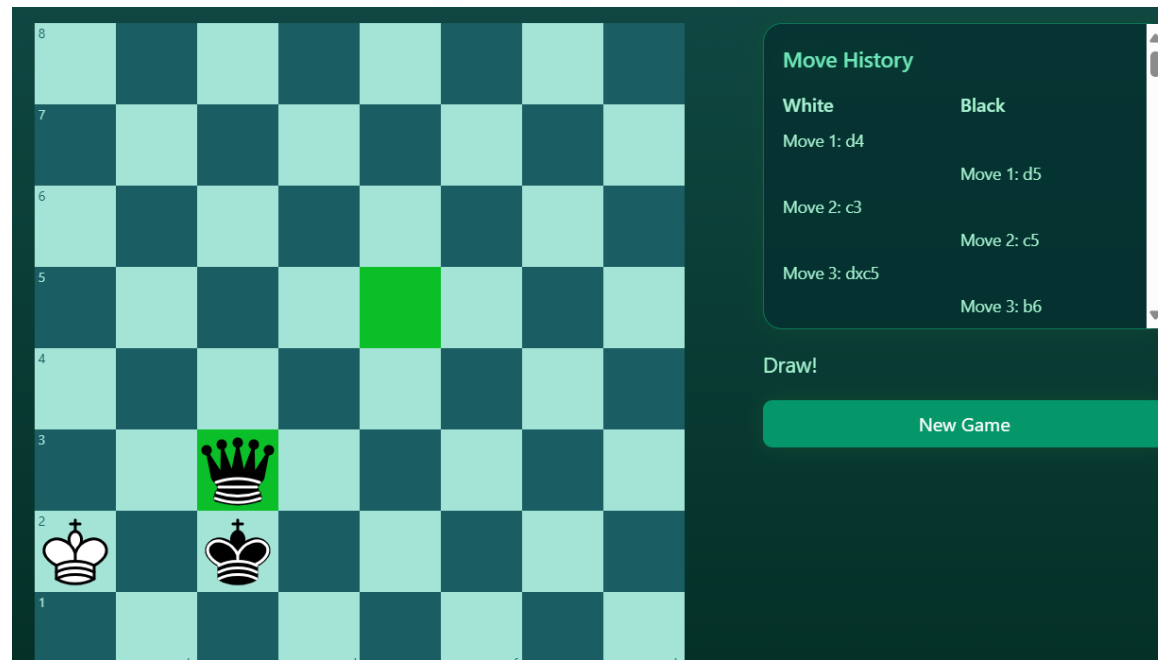


Figure 9: Chess draw

Draw by 3-times constant

In a game of chess, if a chess position is moved repeatedly 3 times in a row, the game is considered a draw. At this time, the player immediately reports to the referee to confirm this draw result.

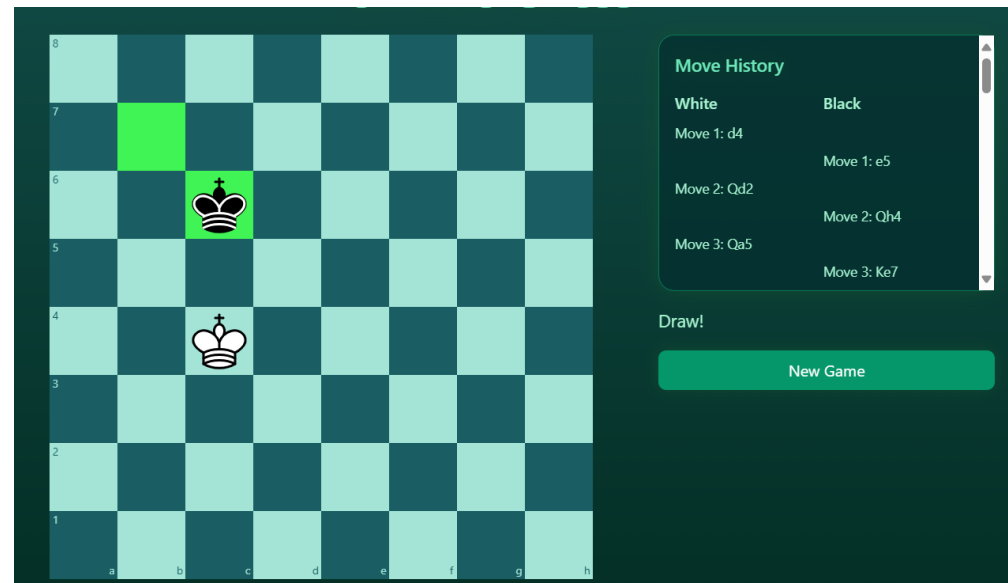


Figure 10: Chess draw 2

Draw after 50 moves

When playing chess, if no piece is captured and no pawn is made in 50 moves, the game is declared a draw. The referee will base on the game record to declare a draw when a player requests to stop the game.

However, the number of 50 moves is increased to 75 moves in the endgame when both sides have the following pieces:

- King + Rook + Bishop against King + Rook
- King + 2 Knights against King + Pawn
- King + Queen against 2 Knights
- King + Queen + Pawn (one square left until the promotion square) against King + Queen
- King + Queen against King + 2 Bishops
- King + 2 Bishops against King + Knight

Special move

Pawn Promotion in Chess

Promotion is a rule in chess that allows a pawn to become a stronger piece when it reaches the end of the board.

If a white pawn reaches the 8th rank, or a black pawn reaches the 1st rank, it must be promoted.

Players can choose one of four pieces:

- Queen
- Rook
- Bishop
- Knight

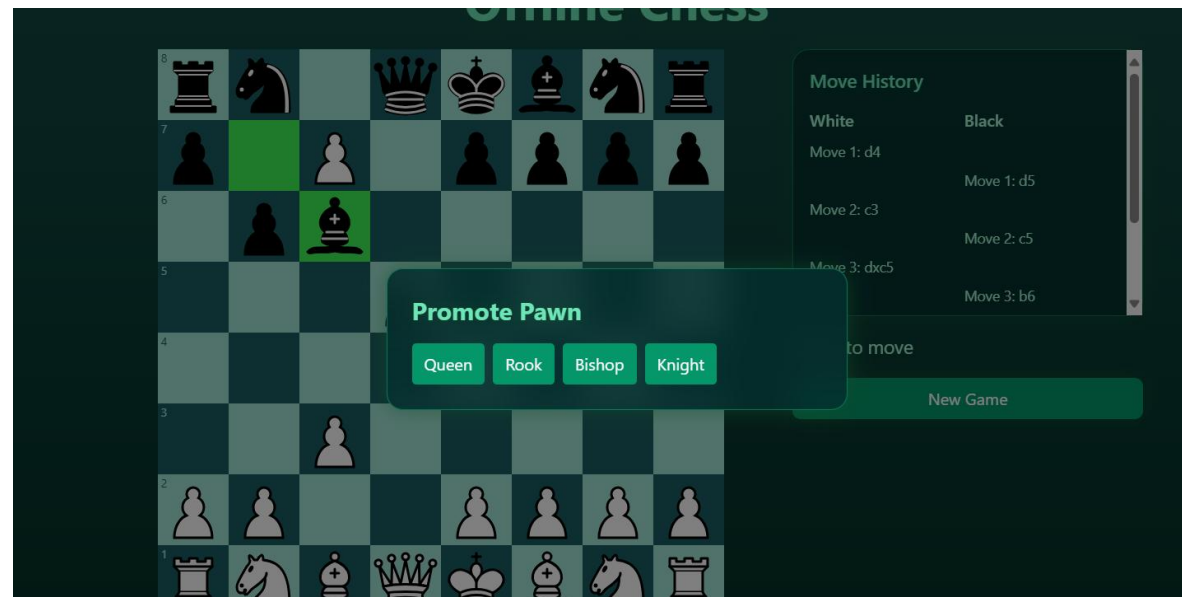


Figure 11: Pawn Promotion in Chess

The pawn has become the chariot.



Figure 12: The pawn has become the chariot.

3.1.5. Player Profile

Player profiles on the online chess platform are designed to enhance the personalized experience and provide detailed information about the performance. Players can track their development, customize their diagrams, and interact with the chess community flexibly.

Match History

Players can access the match history to review previous games, including wins, losses, draws, and important moves.

The system displays detailed, limited statistics such as win rate, longest winning streak, average time in each battle, helping players analyze tactics and improve their skills.

Players can review the moves of each battle, supporting the learning process and studying strengths or mistakes to gain experience.

Custom Profile

Players can choose an avatar, set a display name, write a short description about themselves, helping to create a personal mark in the community.

Chess ownership can be edited, displaying favorite playing styles (e.g. attack, defense, tool tactics).

This feature helps build connections between players, creating a more interesting competitive and trading environment.

3.2. Required functionality

3.2.1. Performance

Optimal performance is an important factor in the online chess experience. The system must be designed to respond quickly to player actions, ensuring a smooth experience.

Page load time: The platform must load in less than 3 seconds on normal internet connections. This enhances convenience and avoids causing an interrupted experience for players.

Input accuracy: When a player makes a move, feedback must appear within 100 milliseconds, ensuring no significant readout. This keeps the game seamless and accurate, especially in rapid chess matches (Blitz, Bullet).

System stability: The server must be able to handle a lot of simultaneous players without affecting the speed or quality of the game. Using powerful servers and optimized algorithms will help maintain stable performance even at peak times. All of the above factors ensure a fast, smooth, lag-free platform, bringing a professional and engaging chess playing experience to every user.

3.2.2. Security

User data security is of utmost importance for the online chess platform. To protect sensitive information, user data will be encrypted using advanced algorithms before being stored. This ensures that personal information, match history and login information are encrypted both at rest and in transit, helping to prevent unauthorized access or cyber attacks.

SSL Encryption: All data transmitted between users and the server will be protected by SSL (Secure Socket Layer) protocol, preventing information theft such as username, password, match history. This helps players feel secure when participating in the platform.

Authentication & Account Protection: The system will apply two-factor authentication (2FA) for login, helping to enhance security. Protection measures against Brute Force, SQL Injection and XSS attacks are also deployed to ensure the platform is not attacked.

Regular security checks: The system will perform regular security checks and updates to patch potential vulnerabilities, ensuring safety against new threats.

3.2.3. User Experience (Usability)

The user interface (UI) of the online chess platform will be designed to be intuitive, simple, and easy to use to ensure that every player can navigate quickly.

Clear, intuitive layout: Function buttons, chessboard, personal profile, and leaderboard will be arranged logically to help users easily access.

Support for new players:

Detailed chess instructions will help new players get acquainted with the rules of the game.

Smart move suggestions (optional on/off) help players better understand tactics.

Support tooltips: When users hover over a function, there will be a short description to help them understand how to use it.

Optimized experience on all devices: The interface will automatically adapt to computer, phone, and tablet screens, ensuring a seamless experience.

3.2.4. Compatibility

The online chess platform will be designed to be compatible with all popular browsers, ensuring that all players can easily access it without display errors.

Browser support: The platform must work smoothly on Google Chrome, Firefox, Microsoft Edge, Safari and other popular browsers.

Responsive Design:

Automatically adjust the interface to suit different screen sizes.

Dark Mode and Light Mode for players to choose the interface according to their preferences.

Optimized mobile experience:

Touch-friendly interface on smartphones and tablets.

Supports swipe and drag operations to move chess pieces.

Optimized performance to ensure no lag even on low-configuration devices.

All of the above factors help ensure the platform operates stably on all devices, providing a professional and convenient online chess playing experience.

IV. External Interface Requirements

4.1. User Interfaces

The online chess platform is designed with an intuitive, easy-to-use interface, helping players easily navigate and experience the game. The interface is divided into the following main sections:

Main Menu:

The navigation center of the platform, providing features such as registration, login, and game mode selection.

The registration process is quick, requiring only basic information so that players can participate immediately.

Choose a variety of game modes, including playing against AI, playing with friends, and participating in ranking tournaments.

Game Interface:

The chess board is clearly displayed, placed in the center of the screen for players to easily observe and come up with strategies.

Displaying time and moves, helping to accurately track the progress of the match.

The scoreboard and rankings are displayed right next to it, helping players track their personal achievements.

Integrate live chat feature, support communication with opponents, discuss tactics or chat for fun.

Profile Page:

Display detailed statistics: number of wins/losses/draws, playing time, ranking, recent performance.

Customize avatar, display name, help personalize the player experience.

Configure settings, allow changing sound, interface, display mode according to preferences.

4.2. Hardware Interfaces

The online chess platform is designed to run smoothly on popular devices without requiring specialized hardware.

Supported on multiple devices:

Desktop (Windows, macOS, Linux)

Laptop (Laptop)

Tablet (iPad, Android Tablet)

Smartphone (iOS, Android)

Minimum hardware requirements:

Intel Core i3 processor or equivalent

Minimum 2GB RAM to ensure lag-free experience

Stable internet connection to ensure continuous data transmission

Optimized for mobile experience:

Smooth touch interface, supporting drag/drop chess pieces.

Flexible screen rotation mode, suitable for both large and small screen devices.

4.3. Software Interfaces

The online chess system is built with a powerful software architecture, ensuring high performance, flexible expansion and smooth experience.

Back-End:

Using Node.js as the main server, helping with multi-threading and fast response.

Player data, match history are stored with MongoDB, ensuring fast and secure retrieval.

Integrating WebSocket to support real-time play, helping players' moves to be updated immediately without reloading the page.

Front-End:

Using React.js/Vue.js to build a flexible interface and optimize performance.

Support Dark Mode and Light Mode to customize the interface according to player preferences.

Integrating CSS animations for a smoother, more vivid experience.

API and Integration:

Support logging in with Google/Facebook, helping players access quickly without having to create a new account.

AI (Stockfish) integration for computer mode, providing worthy challenges for players. With this optimized software system, the online chess platform will ensure stable performance, high scalability and the best user experience.



Figure 13:MongoDB



Figure 14:IntelliJ IDEA Community Edition

Front-End: On the front-end, the website will be crafted using a combination of HTML, CSS, and modern JavaScript frameworks, such as React or Vue.js. These frameworks are known for their ability to create highly interactive and dynamic user interfaces, which are essential for engaging players in real-time gameplay. React, for instance, allows for the development of reusable components, which can streamline the coding process and improve maintainability, while Vue.js offers a more flexible approach with a gentle learning curve for developers. Utilizing these frameworks will enable the site to provide a responsive and apparently appealing experience, as users can interact with the game seamlessly, receiving immediate feedback for their actions. Furthermore, the integration of CSS will ensure that the design is not only functional but also aesthetically pleasing, enhancing the overall user experience.



Figure 15:Visua studio Code

V.Usecase

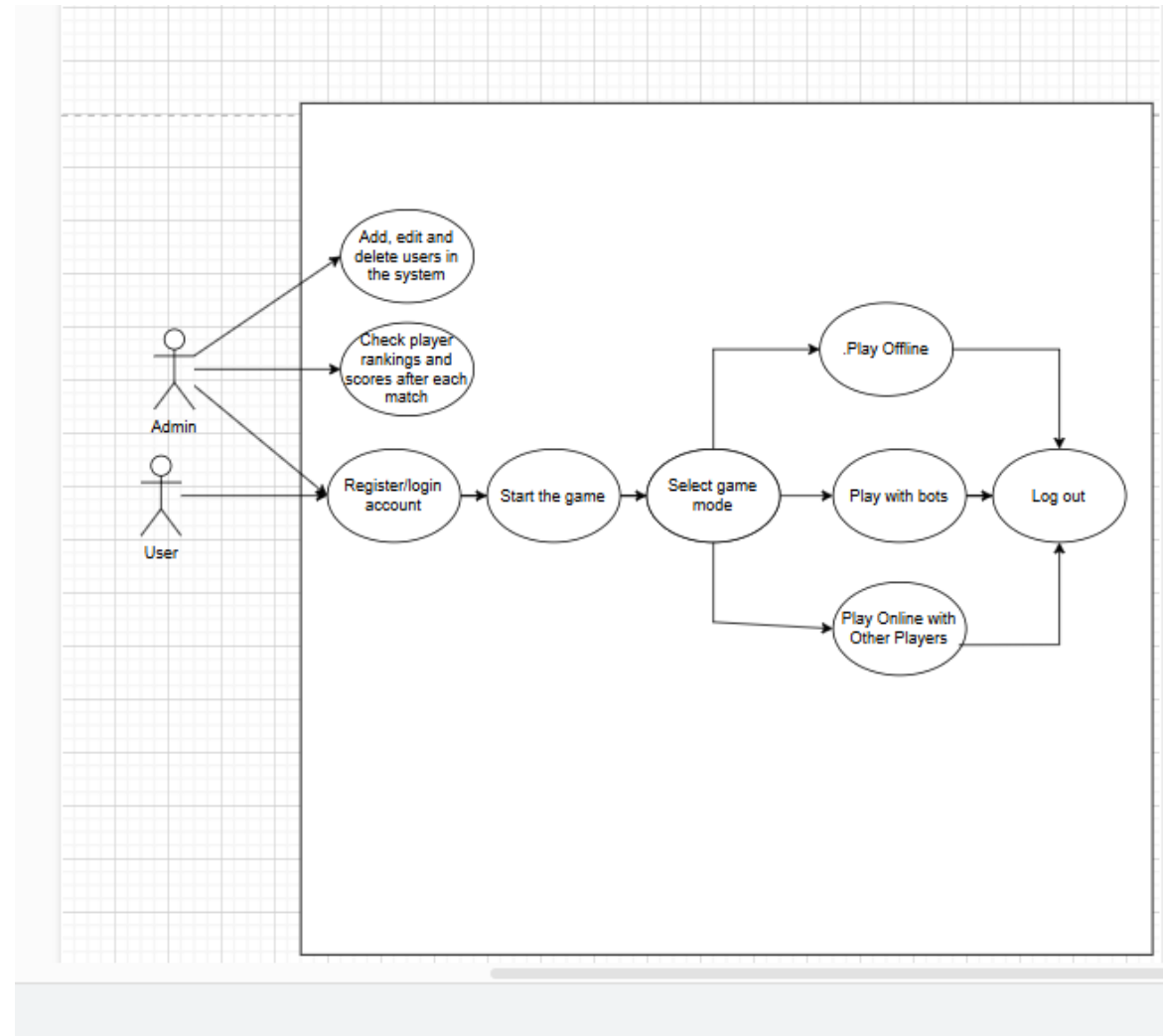


Figure 16:Usecase

Below is an edited version of the Chess Website system description to match the Use Case Diagram you provided.

1. Actors Description

a. User

Description:

Chess players can perform basic functions such as registration, selecting game mode and logging out.

Main actions:

Register/Login account: Authenticate before joining the system.

Start game: Open the chess board to prepare for the match.

Choose game mode: Choose the appropriate game format.

Play offline.

Play with bots.

Play online with other players.

Log out of the system after completing the matches.

b. Administrator

Description:

Administrators have control over the system, mainly managing player accounts and tracking rankings after each match.

Main Actions:

Add, edit and delete users in the system.

Check the player's ranking and score after each match.

2. Main Use Cases

A. User Functions

2.1. Register/Login account

Purpose: Authenticate users before accessing the system.

Details:

Register: User enters personal information (email, password) → Verify email.

Login: System checks information → Grant access.

2.2. Start the game

Purpose: Open the board and prepare for competition.

Activate: When the user clicks the "Start Game" button.

3. Select game mode

Options:

Play Offline: Play without an Internet connection.

Play with bots: Compete with AI (machine) at different levels.

Play Online with Other Players: Play with other players (random or friends).

4. Use Cases Description

#	Code	Name	Brief Description
1	UC01	Register/Login	Allows actors to register/login to the system.
2	UC02	modify game settings	Allows admin to adjust parameters such as: game rules, difficulty level or server settings
3	UC03	Play game	Allows players to join a match.
4	UC04	Choose game mode	Allows players to choose game mode
5	UC05	Play offline	Allow users to play offline
6	UC06	Play online	Allows players to play online.
7	UC07	Play with AI	Allows players to play games with AI
8	UC08	View leaderboard	Allows players to check the rankings of top players.
9	UC09	View player	Allows players to browse player profiles and stats.

5. Use Case & Actor mapping

Use case \ Actor	Admin	User (Player)
UC01: Resgiter/ Login	x	x
UC02: modify game settings		x
UC03: Play game	x	
UC04: Choose game mode	x	
UC05 :Play offline		x
UC06: Play online	x	x
UC07: Play with AI	x	x
UC08: View leaderboard	x	x
UC09: View player	x	x